> 📚 참고: 이 실습은『파이썬 라이브러리를 활용한 머신러닝』(안드레아스 뮐러·세라 가이도 지음, 박해선 옮김, 한빛미디어) 예제와 저자의 `mglearn` 패키지(BSD 라이선스)를 바탕으로 합니다.

> 🛠️ **2026.10 수정본 (v11)** — 원본: ch07_02_movieReview.ipynb
>
> 데이터가 없으면 스탠퍼드 원본에서 IMDB 리뷰(84MB)를 내려받아 `data/aclImdb` 에 푸는 셀을 추가했어요(이미 있으면 건너뜀). 레이블 없는 `train/unsup` (5만 개)은 빼고 `neg` · `pos` 두 클래스만 읽도록 했어요 — 그래서 훈련 데이터는 75000개가 아니라 **25000개**예요. `get_feature_names()` → `get_feature_names_out()` (scikit-learn 1.2+), 마지막 히트맵은 최신 matplotlib에서 `mglearn.tools.heatmap` 이 오류가 나서 seaborn으로 그렸어요. 본문의 행렬 크기(75000 x 124255)를 실제 실행 결과(25000 x 74849)로 고쳤고, 파이프라인 그리드 서치는 `n_jobs=-1` 로 CPU를 모두 쓰게 했어요.

영화 리뷰 감성 분석

학습 내용

In [ ]:

- 데이터 셋 : 스탠퍼드 대학교 연구원 앤드류 마스가IMDB(Internet Movie Database)사이트에서 수집한 영화 리뷰 데이터 셋
- http://ai.stanford.edu/~amaas/data/sentiment/
- 리뷰 텍스트는 양성(긍정 평가), 음성(부정 평가)의 레이블 포함.
- 점수 : 7점 이상은 양성, 4점 이하는 음성인 이진 분류 데이터
- 다운로드 후, 압축 해제후, 레이블이 되어 있지 않은 폴더는 삭제

In [1]:
```
import numpy as np
```

01 데이터 불러오기

데이터 내려받기 (2026.10 추가)

- `data/aclImdb` 폴더가 없으면 스탠퍼드 원본 사이트에서 내려받아 풉니다. 이미 있으면 건너뜁니다.

In [2]:
```
# 2026.10 추가: IMDB 영화 리뷰 데이터 내려받기 (이미 있으면 건너뜀, 윈도우·맥·코랩 공통)
import os, tarfile, urllib.request

url = "https://ai.stanford.edu/~amaas/data/sentiment/aclImdb_v1.tar.gz"   #
os.makedirs("data", exist_ok=True)
```

```
tar_path = os.path.join("data", "aclImdb_v1.tar.gz")

if os.path.isdir(os.path.join("data", "aclImdb", "train")):
    print("data/aclImdb 폴더가 이미 있어 내려받기를 건너뜁니다.")
else:
    if not os.path.exists(tar_path):
        print("내려받는 중... (84MB, 몇 분 걸릴 수 있어요)")
        urllib.request.urlretrieve(url, tar_path)
    print("압축 푸는 중...")
    with tarfile.open(tar_path, "r:gz") as tar:
        # 레이블이 없는 train/unsup(5만 개)은 감성 분석에 쓰지 않으므로 풀지 않습니다.
        members = [m for m in tar.getmembers() if "/unsup/" not in m.name ar
        opt = {"filter": "data"} if hasattr(tarfile, "data_filter") else {}
        tar.extractall("data", members=members, **opt)
    print("완료:", os.listdir(os.path.join("data", "aclImdb")))
```

```
data/aclImdb 폴더가 이미 있어 내려받기를 건너뜁니다.
```

In [3]:
```
from sklearn.datasets import load_files

reviews_train = load_files("data/aclImdb/train/", categories=["neg", "pos"])

text_train, y_train = reviews_train.data, reviews_train.target

# 텍스트와 레이블을 포함하고 있는 Bunch 오브젝트를 반환
text_train, y_train = reviews_train.data, reviews_train.target
print("text_train의 타입: {}".format(type(text_train)))
print("text_train의 길이: {}".format(len(text_train)))
print("text_train[6]:\n{}".format(text_train[6]))
```

```
text_train의 타입: <class 'list'>
text_train의 길이: 25000
text_train[6]:
b"This movie has a special way of telling the story, at first i found it rat
her odd as it jumped through time and I had no idea whats happening.<br /><b
r />Anyway the story line was although simple, but still very real and touch
ing. You met someone the first time, you fell in love completely, but broke
up at last and promoted a deadly agony. Who hasn't go through this? but we w
ill never forget this kind of pain in our life. <br /><br />I would say i am
rather touched as two actor has shown great performance in showing the love
between the characters. I just wish that the story could be a happy ending."
```

In [4]:
```
# 줄바꿈 문자 삭제
text_train = [doc.replace(b"<br />", b" ") for doc in text_train]
```

In [5]:
```
# 양성과 음성의 샘플이 같다.
print("클래스별 샘플 수 (훈련 데이터): {}".format(np.bincount(y_train)))
```

```
클래스별 샘플 수 (훈련 데이터): [12500 12500]
```

테스트 데이터 불러오기

In [6]:
```
reviews_test = load_files("data/aclImdb/test/")
text_test, y_test = reviews_test.data, reviews_test.target
print("테스트 데이터의 문서 수: {}".format(len(text_test)))
```

```
print("클래스별 샘플 수 (테스트 데이터): {}".format(np.bincount(y_test)))
text_test = [doc.replace(b"<br />", b" ") for doc in text_test]
```

```
테스트 데이터의 문서 수: 25000
클래스별 샘플 수 (테스트 데이터): [12500 12500]
```

In [ ]:

데이터 전처리

- 텍스트 데이터를 머신러닝 알고리즘에 적용이 가능하도록 수치 표현으로 변경

02 텍스트 데이터를 BOW(bag of word)로 표현

- 장, 문단, 문장, 서식 같은 입력 텍스트의 구조는 잃고 각 단어가 말뭉치에 있는 텍스트에 얼마나 많이 나타나는가 확인

샘플 데이터로 확인

- BOW 표현은 CountVectorizer의 변환기 인터페이스로 구현되어 있음.

In [7]:
```
bards_words =["The fool doth think he is wise,",
              "but the wise man knows himself to be a fool"]
```

- CountVectorizer의 fit메서드는 훈련 데이터를 토큰으로 나누고 어휘 사전을 구축하여 vocabulary_ 속성에 저장

In [8]:
```
from sklearn.feature_extraction.text import CountVectorizer
vect = CountVectorizer()
vect.fit(bards_words)
```

Out[8]:

▾ CountVectorizer ⓘ ⓘ

▸ Parameters

▸ Fitted attributes

▸ 13 features

- 어휘 사전은 'be'에서 'wise'까지 13개의 단어로 구성되어 있음.

In [9]:
```
print("어휘 사전의 크기: {}".format(len(vect.vocabulary_)))
print("어휘 사전의 내용:\n {}".format(vect.vocabulary_))
```

```
어휘 사전의 크기: 13
어휘 사전의 내용:
 {'the': 9, 'fool': 3, 'doth': 2, 'think': 10, 'he': 4, 'is': 6, 'wise': 12,
'but': 1, 'man': 8, 'knows': 7, 'himself': 5, 'to': 11, 'be': 0}
```

훈련 데이터에 대해 BOW 표현을 만들려면 transform 메서드를 호출

In [10]:
```
bag_of_words = vect.transform(bards_words)
print("BOW: {}".format(repr(bag_of_words)))
```

```
BOW: <Compressed Sparse Row sparse matrix of dtype 'int64'
        with 16 stored elements and shape (2, 13)>
```

In [11]:
```
print("BOW의 밀집 표현:\n{}".format(bag_of_words.toarray()))
```

```
BOW의 밀집 표현:
[[0 0 1 1 1 0 1 0 0 1 1 0 1]
 [1 1 0 1 0 1 0 1 1 1 0 1 1]]
```

- BOW 표현은 0이 아닌 값만 저장하는 SciPy 희소 행렬로 저장되어 있음.
- 행렬의 크기는 2 x 13, 각각의 행은 하나의 데이터 포인터를 나타낸다.
- 각 특성은 어휘 사전에 있는 각 단어에 대응.

03 영화 리뷰에 대한 BOW

In [12]:
```
vect = CountVectorizer().fit(text_train)
X_train = vect.transform(text_train)
print("X_train:\n{}".format(repr(X_train)))
```

```
X_train:
<Compressed Sparse Row sparse matrix of dtype 'int64'
        with 3431196 stored elements and shape (25000, 74849)>
```

- 훈련 데이터의 BOW 표현 X_train의 크기는 25000 x 74849 (2026.10 실행 기준, `unsup` 제외)
- 어휘사전은 단어를 74849개를 담고 있음.
- 데이터는 Scipy희소 행렬로 저장되어 있음.

In [13]:
```
feature_names = vect.get_feature_names_out()  # 2026.10 수정: get_feature_nam
print("특성 개수: {}".format(len(feature_names)))
```

```
특성 개수: 74849
```

In [14]:
```
# 처음 15개가 숫자임. 리뷰 어딘가에 나타남.
print("처음 20개 특성:\n{}".format(feature_names[:20]))
```

```
처음 20개 특성:
['00' '000' '0000000000001' '00001' '00015' '000s' '001' '003830' '006'
 '007' '0079' '0080' '0083' '0093638' '00am' '00pm' '00s' '01' '01pm' '02']
```

In [15]:
```
print("20010에서 20030까지 특성:\n{}".format(feature_names[20010:20030]))
```

```
20010에서 20030까지 특성:
['dratted' 'draub' 'draught' 'draughts' 'draughtswoman' 'draw' 'drawback'
 'drawbacks' 'drawer' 'drawers' 'drawing' 'drawings' 'drawl' 'drawled'
 'drawling' 'drawn' 'draws' 'draza' 'dre' 'drea']
```

In [16]:
```
print("매 2000번째 특성:\n{}".format(feature_names[::2000]))
```

```
매 2000번째 특성:
['00' 'aesir' 'aquarian' 'barking' 'blustering' 'bête' 'chicanery'
 'condensing' 'cunning' 'detox' 'draper' 'enshrined' 'favorit' 'freezer'
 'goldman' 'hasan' 'huitieme' 'intelligible' 'kantrowitz' 'lawful' 'maars'
 'megalunged' 'mostey' 'norrland' 'padilla' 'pincher' 'promisingly'
 'receptionist' 'rivals' 'schnaas' 'shunning' 'sparse' 'subset'
 'temptations' 'treatises' 'unproven' 'walkman' 'xylophonist']
```

In [17]:
```
# X_train = vect.transform(text_train)
print("X_train:\n{}".format(repr(X_train)))
```

```
X_train:
<Compressed Sparse Row sparse matrix of dtype 'int64'
        with 3431196 stored elements and shape (25000, 74849)>
```

In [18]:
```
from sklearn.model_selection import cross_val_score
from sklearn.linear_model import LogisticRegression
```

In [19]:
```
scores = cross_val_score(LogisticRegression(), X_train, y_train, cv=5)
print("크로스 밸리데이션 평균 점수: {:.2f}".format(np.mean(scores)))
```

```
.../site-packages/sklearn/linear_model/_logistic.py:599: ConvergenceWarning:
lbfgs failed to converge after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regre
ssion
  n_iter_i = _check_optimize_result(
.../site-packages/sklearn/linear_model/_logistic.py:599: ConvergenceWarning:
lbfgs failed to converge after 100 iteration(s) (status=1):
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ssion
  n_iter_i = _check_optimize_result(
```

```
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ssion
  n_iter_i = _check_optimize_result(
```

크로스 밸리데이션 평균 점수: 0.88

```
.../site-packages/sklearn/linear_model/_logistic.py:599: ConvergenceWarning:
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ssion
  n_iter_i = _check_optimize_result(
```

- 교차 검증 평균 점수를 88%를 얻음.

매개변수 C를 이용하여 그리드 서치를 사용하여 조정

In [20]:
```python
from sklearn.model_selection import GridSearchCV
param_grid = {'C': [0.001, 0.01, 0.1, 1, 10]}

# LogisticRegression() 일경우 WARNING메세지 발생
grid = GridSearchCV(LogisticRegression(max_iter=500), param_grid, cv=5)
grid.fit(X_train, y_train)
print("최상의 크로스 밸리데이션 점수: {:.2f}".format(grid.best_score_))
print("최적의 매개변수: ", grid.best_params_)
```

```
최상의 크로스 밸리데이션 점수: 0.89
최적의 매개변수:  {'C': 0.1}
```

In [21]:
```python
X_test = vect.transform(text_test)
print("테스트 점수: {:.2f}".format(grid.score(X_test, y_test)))
```

```
테스트 점수: 0.88
```

In [22]:
```
vect = CountVectorizer(min_df=5).fit(text_train)
X_train = vect.transform(text_train)
print("min_df로 제한한 X_train: {}".format(repr(X_train)))
```

```
min_df로 제한한 X_train: <Compressed Sparse Row sparse matrix of dtype 'int64'
        with 3354014 stored elements and shape (25000, 27271)>
```

In [23]:
```
feature_names = vect.get_feature_names_out()  # 2026.10 수정: get_feature_nam

print("First 50 features:\n{}".format(feature_names[:50]))
print("Features 20010 to 20030:\n{}".format(feature_names[20010:20030]))
print("Every 700th feature:\n{}".format(feature_names[::700]))
```

```
First 50 features:
['00' '000' '007' '00s' '01' '02' '03' '04' '05' '06' '07' '08' '09' '10'
 '100' '1000' '100th' '101' '102' '103' '104' '105' '107' '108' '10s'
 '10th' '11' '110' '112' '116' '117' '11th' '12' '120' '12th' '13' '135'
 '13th' '14' '140' '14th' '15' '150' '15th' '16' '160' '1600' '16mm' '16s'
 '16th']
Features 20010 to 20030:
['repentance' 'repercussions' 'repertoire' 'repetition' 'repetitions'
 'repetitious' 'repetitive' 'rephrase' 'replace' 'replaced' 'replacement'
 'replaces' 'replacing' 'replay' 'replayable' 'replayed' 'replaying'
 'replays' 'replete' 'replica']
Every 700th feature:
['00' 'affections' 'appropriately' 'barbra' 'blurbs' 'butchered' 'cheese'
 'commitment' 'courts' 'deconstructed' 'disgraceful' 'dvds' 'eschews'
 'fell' 'freezer' 'goriest' 'hauser' 'hungary' 'insinuate' 'juggle'
 'leering' 'maelstrom' 'messiah' 'music' 'occasional' 'parking'
 'pleasantville' 'pronunciation' 'recipient' 'reviews' 'sas' 'shea'
 'sneers' 'steiger' 'swastika' 'thrusting' 'tvs' 'vampyre' 'westerns']
```

In [24]:
```
grid = GridSearchCV(LogisticRegression(), param_grid, cv=5)
grid.fit(X_train, y_train)
print("최적의 크로스 밸리데이션 점수: {:.2f}".format(grid.best_score_))
```

```
.../site-packages/sklearn/linear_model/_logistic.py:599: ConvergenceWarning:
lbfgs failed to converge after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).
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ssion
  n_iter_i = _check_optimize_result(
```

```
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```

```
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```

최적의 크로스 밸리데이션 점수: 0.89

```
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ssion
  n_iter_i = _check_optimize_result(
```

불용어

- 유용하지 않은 단어를 제거하기
  - (1) 언어별 불용어(stopword)목록을 사용하기
    - feature_extraction.text 모듈에 영어의 불용어 사용
  - (2) 너무 자주 나타나는 단어를 제외

In [25]:
```
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS
print("불용어 개수: {}".format(len(ENGLISH_STOP_WORDS)))
print("매 10번째 불용어:\n{}".format(list(ENGLISH_STOP_WORDS)[::10]))
```

```
불용어 개수: 318
매 10번째 불용어:
['wherever', 'down', 'least', 'fifteen', 'so', 'own', 'thin', 'nine', 'somet
ime', 'whereas', 'detail', 'herein', 'her', 'call', 'again', 'this', 'thos
e', 'these', 'top', 'ours', 'while', 'put', 'why', 'interest', 'hundred', 'a
lready', 'mill', 'cry', 'among', 'should', 'somehow', 'show']
```

In [26]:
```
# stop_words="english"라고 지정하면 내장된 불용어를 사용합니다.
# 내장된 불용어에 추가할 수도 있고 자신만의 목록을 사용할 수도 있습니다.
vect = CountVectorizer(min_df=5, stop_words="english").fit(text_train)
X_train = vect.transform(text_train)
print("불용어가 제거된 X_train:\n{}".format(repr(X_train)))
```

```
불용어가 제거된 X_train:
<Compressed Sparse Row sparse matrix of dtype 'int64'
        with 2149958 stored elements and shape (25000, 26966)>
```

In [27]:
```
grid = GridSearchCV(LogisticRegression(), param_grid, cv=5)
grid.fit(X_train, y_train)
print("최상의 크로스 밸리데이션 점수: {:.2f}".format(grid.best_score_))
```

```
.../site-packages/sklearn/linear_model/_logistic.py:599: ConvergenceWarning:
lbfgs failed to converge after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

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ssion
  n_iter_i = _check_optimize_result(
```

최상의 크로스 밸리데이션 점수: 0.88

In [28]:

```
from sklearn.pipeline import make_pipeline
pipe = make_pipeline(CountVectorizer(), LogisticRegression())
param_grid = {'countvectorizer__max_df': [100, 1000, 10000, 20000], 'logisti
grid = GridSearchCV(pipe, param_grid, cv=5, n_jobs=-1)  # 2026.10 수정: 100번
grid.fit(text_train, y_train)
print("최상의 크로스 밸리데이션 점수: {:.2f}".format(grid.best_score_))
print(grid.best_params_)
```

```
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Increase the number of iterations to improve the convergence (max_iter=100).
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Please also refer to the documentation for alternative solver options:
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.../site-packages/sklearn/linear_model/_logistic.py:599: ConvergenceWarning:
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  n_iter_i = _check_optimize_result(
```

```
최상의 크로스 밸리데이션 점수: 0.89
{'countvectorizer__max_df': 20000, 'logisticregression__C': 0.1}
```

```
.../site-packages/sklearn/linear_model/_logistic.py:599: ConvergenceWarning:
lbfgs failed to converge after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

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ssion
  n_iter_i = _check_optimize_result(
```

In [29]:
```
len(grid.best_estimator_.named_steps['countvectorizer'].vocabulary_)
```

Out[29]:
```
74840
```

In [30]:
```
# 2026.10 수정: mglearn.tools.heatmap 은 최신 matplotlib 에서 TypeError 가 나서 s
#               (행 = max_df 4개, 열 = C 5개가 되도록 .T 없이 reshape)
import seaborn as sns
import matplotlib.pyplot as plt

scores = grid.cv_results_['mean_test_score'].reshape(-1, 5)
# 히트맵을 그립니다
sns.heatmap(scores, annot=True, fmt=".3f", cmap="viridis",
            xticklabels=param_grid['logisticregression__C'],
            yticklabels=param_grid['countvectorizer__max_df'])
plt.xlabel("C")
plt.ylabel("max_df")
plt.show()
```

```
/var/folders/7h/rt3bmzzd3dg9wmwz7g3s98jc0000gn/T/ipykernel_61173/162487024.p
y:13: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be sh
own
  plt.show()
```

In [ ]:
```
```